CAN-FD Schematic Byte-wise Design

Secure Segmented CAN-FD Architecture — Phase 1 Frame Specification

Version 1.6 | Capstone Project — 7th Semester | August 09, 2026
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1. Executive Summary

This document specifies the complete byte-level layout of every CAN-FD frame transmitted in Phase 1 of the Capstone project. The design targets a 4-domain segmented vehicle network secured by post-quantum cryptography (Ascon-AEAD128) and hardware-rooted identity (SRAM PUF + ML-DSA-44 / FIPS 204). Primary payload frames (Category 1 Telemetry, Category 2 Attestation, Category 3.1 Challenge, Category 3.3 Key Delivery) use full 64-byte CAN-FD payloads (DLC 15). Control NACK frames (Category 3.2) use compact 16-byte payloads (DLC 10). The Central Intelligent Security Gateway (CISG) acts as the sole root of trust and the single decryption/re-encryption hub under a Pure Star Topology.

> **NOTE** Revision v1.5 (Frozen Wire & Crypto Specification): Resolves all 12 wire/crypto protocol discrepancies: (1) Standardized on ML-DSA-44 (FIPS 204 / Dilithium2); (2) Whole-object AEAD protection of 2420B signature -> 2436B protected object -> 41 x 60B transport frames; (3) One fresh 16B nonce per attestation transfer; retransmissions resend original ciphertext fragment; (4) Frozen key lifecycle: K_boot_i from pre-hashed PUF + challenge; CISG generates fresh random 128-bit K_i post-attestation; (5) Category 3.2 NACK is boot-time Category-2-only; Category 1 telemetry is NEVER retransmitted; (6) Category 2 payload renamed to 'fragment of the Ascon-AEAD128 protected attestation object'.

> **CAUTION** CITATION INTEGRITY: AI-sourced paper references (MaCAN, SSKT, NSSK-CAN, J1939-ADBE, specific quote numbers) are UNVERIFIED LEADS. Verify each DOI and quote against the primary source before including in the final report. The engineering rationale for Pure Star is independently valid and does not require any of these citations.

2. Architecture Overview

2.1 Network Topology

Four electrically isolated CAN-FD domains are interconnected exclusively through the CISG. All inter-domain traffic and all crypto operations are mediated by the gateway.

Domain	Domain ID	Representative ECUs	Bus Speed
Powertrain	0x01	Engine ECU (ECM), Transmission, Throttle Control	2 Mbps data phase
Chassis	0x02	EPS, ESC, Suspension, Brake Controller / ABS	2 Mbps data phase
Body & Comfort	0x03	BCM, Climate, Lighting, Door/Window Control	500 kbps
Infotainment	0x04	Head Unit, Telematics, GPS	500 kbps

2.2 Key Architecture & Handshake Lifecycle — Pure Star Topology (Phase 1)

Every ECU holds exactly one runtime session key K_i. K_i is randomly generated by the CISG after successful ML-DSA attestation and securely delivered to the ECU using K_boot_i. The CISG holds the enrollment table {ECU_ID -> enrolled PUF_response_hash, ML-DSA public key}. Before K_i exists, boot-time attestation frames (Category 2 & 3.2) and key delivery (Category 3.3) are encrypted under an Ephemeral Bootstrap Key K_boot_i derived immediately after Challenge.

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BOOT HANDSHAKE LIFECYCLE:
  1. CISG -> ECU_i : Category 3.1 Challenge (Plaintext)
                     Transmits RANDOM_CHALLENGE (16B) & mints SESSION_ID (1B).

  2. BOTH DERIVE BOOTSTRAP KEY (K_boot_i):
                     ECU_i computes PUF_response_hash = SHA-256(PUF_response).
                     PRK_boot = HKDF-Extract(salt = RANDOM_CHALLENGE, IKM =
PUF_response_hash)
                     K_boot_i = HKDF-Expand(PRK_boot, info = "CAPSTONE-V1/BOOTSTRAP-KEY" ||
TARGET_ECU_ID || SESSION_ID, L = 16 bytes)

  3. ECU_i -> CISG : Category 2 Attestation Transfer (Encrypted under K_boot_i)
                     Whole-object AEAD protection of 2420B ML-DSA-44 signature -> 2436B
protected object.
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                    Kaustuv's 60/4 protocol fragments 2436B into 41 CAN-FD frames.
                    Selective retransmissions (if any) use Category 3.2 NACK (Plaintext).

  4. CISG -> ECU_i : Category 3.3 Session Key Delivery (Encrypted under K_boot_i)
                    CISG verifies signature. If valid, CISG generates a FRESH RANDOM 128-
bit K_i.
                    Delivers fresh K_i wrapped with Ascon-AEAD128 under K_boot_i.

RUNTIME TOPOLOGY (Pure Star -- every frame transits CISG):
  ECU_i  --[K_i, Ascon-AEAD128]-->  CISG
         decrypt | verify tag | IDS inspect | re-encrypt [K_j]
  CISG   --[K_j, Ascon-AEAD128]-->  ECU_j

  CISG key lookup: always SENDER_ECU_ID (Byte 1) -> K_i
  DOMAIN_ID  (Byte 2): routing + IDS metadata only -- NOT crypto-relevant
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Key	Scope	Derivation / Source	Encryption Target
K_boot_i (Bootstrap Key)	Ephemeral (Boot)	HKDF-Extract(salt=RANDOM_CHALLENGE, IKM=SHA-256(PUF_response))	Cat 2 Attestation, Cat 3.3 Key Delivery
K_i (Runtime Session Key)	Session (Runtime)	Fresh random 128-bit key generated by CISG post-attestation	Cat 1 Protected Runtime Telemetry

NOTE Masquerade Residual: A compromised ECU can forge traffic only under its own K_i. Blast radius is bounded to one ECU's channel. This residual is accepted and documented in the Trust Boundary Analysis; it is mitigated by the Phase 2 LSTM IDS at the CISG.

WARNING Known Phase 1 Simplification — Same-Domain Multi-Consumer Bottleneck: Under pure star, same-domain multi-consumer signals (e.g. ECM speed consumed by ABS and EPS, both in Powertrain domain) require the CISG to decrypt once then re-encrypt separately for each consumer and re-inject onto the same bus. This extends the cross-domain gateway overhead into intra-domain traffic. ACCEPTED for Phase 1: the testbed runs on Docker/SocketCAN virtual interfaces with no real-time constraints. Phase 4 HIL validation on MCP2518FD hardware is the correct measurement point for this cost.

2.3 Cryptographic Parameters

Parameter	Value	Notes
Algorithm	Ascon-AEAD128	NIST SP 800-232; lightweight AEAD standard
Key sizes	128 bits (16 B)	Bootstrap key K_boot_i & Session key K_i
Nonce size	128 bits (16 B)	Constructed per frame category (Cat 1, Cat 2, Cat 3.2, Cat 3.3)
Auth tag	128 bits (16 B)	Full tag -- NO truncation
Associated Data	Category-specific	Category-specific unencrypted control/header fields are authenticated where defined; Category 2 transport metadata is not part of the whole-object AEAD operation.
Key delivery wrap	Ascon-AEAD128	K_i wrapped under K_boot_i = HKDF-Extract(salt=RANDOM_CHALLENGE, IKM=PUF_hash)
Attestation sig	ML-DSA-44 (FIPS 204)	Post-quantum digital signature (~2420B / 41 frames). Offline enrollment.
Identity anchor	SRAM PUF	Hardware root of trust; SHA-256 hashed response enrolled at CISG secure enrollment database

3. Frame Category Definitions

Primary frames (Cat 1, 2, 3.1, 3.3) are 64 bytes (CAN-FD DLC 15). Control NACK (Cat 3.2) is 16 bytes (DLC 10). Byte indices are 0-based, big-endian.

3.1 Category 1: Protected Runtime Telemetry Frame (64 Bytes)

Carries encrypted sensor data during normal vehicle operation. Generated at periodic intervals (typically 10 ms for safety-critical signals). The CISG decrypts using K_sender (looked up by SENDER_ECU_ID, Byte 1), verifies the 16-byte tag, extracts the 12D IDS feature vector, re-encrypts under K_consumer, and forwards.

NOTE Nonce construction for Category 1 (16 bytes = 128-bit Ascon requirement): SENDER_ECU_ID(1B) || DOMAIN_ID(1B) || FLAGS(1B) || 0x00(1B) || MONOTONIC_COUNTER(4B) || 0x00x8. Safety: MONOTONIC_COUNTER never repeats within a session; SENDER_ECU_ID is unique per ECU; K_i is session-scoped via HKDF, so nonces are globally unique across sessions without needing a separate SESSION_ID in the runtime frame header.

Byte Range	Field Name	Size	Description	AD?
Byte 0	PROTOCOL_VER_TYPE	1 B	Bits 7-4: Protocol Version (0x1). Bits 3-0: Message Type (0x1 = Telemetry)	Yes
Byte 1	SENDER_ECU_ID	1 B	Unique 8-bit ECU identifier. PRIMARY CISG key selector -> K_i lookup	Yes
Byte 2	DOMAIN_ID	1 B	0x01=Powertrain 0x02=Chassis 0x03=Body 0x04=Infotainment. Routing + IDS metadata only -- not crypto-relevant in Pure Star	Yes
Byte 3	FLAGS	1 B	Bit 7: Emergency. Bit 6: Encrypted (must be 1). Bit 5: RESERVED (KEY_TIER dropped in v1.2). Bits 4..0: RESERVED	Yes
Bytes 4-7	MONOTONIC_COUNTER	4 B	32-bit Big-Endian counter. Increments per frame per ECU. Resets on re-attestation. At 10 ms cadence: ~497-day rollover.	Yes
Bytes 8-47	ENCRYPTED_PAYLOAD	40 B	Ascon-AEAD128 ciphertext. Plaintext: 7x float32 (28B) + 2x uint16 (4B) + 1x int16 (2B) + 1x uint16 (2B) + 1x uint32 (4B) = 40B	No
Bytes 48-63	ASCON_AUTH_TAG	16 B	128-bit authentication tag -- full length, no truncation	N/A

Telemetry Payload Signal Layout (Bytes 8–47, plaintext before encryption)

Offset in Payload	Field	Type	Unit / Range	IDS Feature?
0–3	Vehicle_Speed	float32	km/h [0–300]	$Signal_1$
4–7	Engine_RPM	float32	rpm [0–8000]	$Signal_2$
8–11	Throttle_Position	float32	% [0–100]	$Signal_3$
12–15	Brake_Pressure	float32	bar [0–200]	$Signal_4$
16–19	Steering_Angle	float32	deg [-540..+540]	$Signal_5$
20–23	Lateral_Accel	float32	g [-5..+5]	$Signal_6$
24–27	Battery_Voltage	float32	V [0–60]	–
28–29	Coolant_Temp	uint16	°C [–40..+215] scaled ×10	–
30–31	Ambient_Temp	uint16	°C [–40..+85] scaled ×10	–
32–33	Gear_Position	int16	–4..+8 (R=–1,N=0,P=–4)	–
34–35	Fault_Code_Count	uint16	count [0–65535]	–
36–39	Odometer	uint32	km × 10 [0..4294967295]	–

3.2 Category 2: 60/4 Indexed Block-Transfer Attestation Frame (64 Bytes)

Used during boot-time to stream the ML-DSA-44 (FIPS 204 / Dilithium2 parameter set) attestation object across exactly 41 frames at 60 bytes/frame. Crypto Layer (Kunal): Encrypts/authenticates the 2420 B signature in a single Ascon-AEAD128 operation using K_boot_i and a single fresh 16B nonce (RANDOM_CHALLENGE), producing a 2436 B protected attestation object (2420 B ciphertext + 16 B tag). Transport Layer (Kaustuv): Fragments the 2436 B object into 41 frames (40 full 60B frames + 1 frame with 36B data + 24B zero-padding).

NOTE Category 2 Nonce & Retransmission Policy: One fresh 16-byte nonce is used for the entire attestation transfer. FRAME_INDEX is transport-layer metadata only. Selective retransmissions (via Cat 3.2 NACK) re-send the exact existing 60-byte ciphertext fragment without re-encryption or new nonces. Attestation Object isThe ECU generates an ML-DSA-44 signature over an attestation transcript containing PROTOCOL_VERSION || ECU_ID || SESSION_ID || RANDOM_CHALLENGE || MEASUREMENT_HASH. RANDOM_CHALLENGE provides freshness and prevents replay of

previously generated attestations, while ECU_ID and SESSION_ID bind the signature to the current ECU and boot session. MEASUREMENT_HASH is the SHA-256 measurement of the approved ECU software/firmware image and security configuration. The raw PUF response is never included in the signed transcript; the PUF is instead used to derive the bootstrap key K_boot_i.

Byte Range	Field Name	Size	Description
Byte 0	SESSION_ID	1 B	8-bit attestation session identifier (minted in Cat 3.1 Challenge).
Byte 1	FRAME_INDEX	1 B	Fragment index 0x00 to 0x28 (0–40). Zero-indexed.
Byte 2	TOTAL_FRAMES	1 B	Always 0x29 (41) for ML-DSA-44 protected attestation object transport.
Byte 3	TRANSFER_FLAGS	1 B	Bits 7–4: Subtype 0x2=BlockData 0x3=Complete 0x4=NACK 0x5=RETRY. Bits 3–0: Retry Counter (max 0xF).
Bytes 4–63	FRAGMENT_PAYLOAD	60 B	60-byte fragment of the Ascon-AEAD128 protected attestation object.

NOTE ML-DSA-44 signature size: ~2420 bytes (Dilithium2 parameter set). 2436 ÷ 60 = 40.6 → 41 frames. The last frame carries 36 bytes of protected data + 24 bytes of zero padding. The public key (~1312 bytes) was written to the CISG secure enrollment database at manufacturing; it is never transmitted on the bus.

3.3 Category 3.1: Gateway Attestation Challenge Frame (64 Bytes, CISG -> ECU)

Sent in plaintext before any session key exists. MINTS the SESSION_ID that all downstream frames (Category 2, 3.2, 3.3) reference. Without SESSION_ID in this frame, downstream frames have no shared session anchor.

Byte Range	Field Name	Size	Description
Byte 0	PROTOCOL_VER_TYPE	1 B	Version 0x1, Msg Type 0x2 (Challenge)
Byte 1	TARGET_ECU_ID	1 B	ID of ECU being challenged
Byte 2	SESSION_ID	1 B	Newly minted 8-bit session identifier. ECU echoes this in all subsequent boot frames (Cat 2, 3.2, 3.3).
Byte 3	ALGO_SELECTION	1 B	0x01 = ML-DSA + Ascon-AEAD128. Note: no Kyber/KEM in this project.
Bytes 4-19	RANDOM_CHALLENGE	16 B	128-bit cryptographically random challenge nonce (CISG TRNG). Also used to derive bootstrap key for Cat 3.3 K_i delivery.
Bytes 20-27	UTC_TIMESTAMP	8 B	64-bit POSIX timestamp -- binds challenge to wall clock
Bytes 28-63	PADDING	36 B	0x00 -- reserved for future protocol extensions

3.4 Category 3.2: NACK / Selective Recovery Frame (16 Bytes, bidirectional)

Used by either side to request or acknowledge retransmission of missing Category 2 fragments. Sent in plaintext (not confidential, not AEAD-protected).

Byte Range	Field Name	Size	Description
Byte 0	SESSION_ID	1 B	Attestation session this NACK belongs to
Byte 1	SENDER_ID	1 B	ECU or CISG issuing the NACK
Byte 2	RETRY_COUNT	1 B	How many retransmission rounds have occurred (max 0xFF)
Byte 3	RESERVED	1 B	0x00
Bytes 4–11	MISSING_BITMAP	8 B	64-bit bitmap: bit N = 1 means frame index N is missing (covers all 41 frames)
Bytes 12–15	RESERVED	4 B	0x00

3.5 Category 3.3: Session Key Delivery Frame (64 Bytes, CISG -> ECU)

Delivers a fresh, randomly generated 128-bit runtime session key K_i after successful ML-DSA-44 attestation. Pure Star: one K_i per ECU -- no domain key, no derived second key. K_i is wrapped with Ascon-AEAD128 under Ephemeral Bootstrap Key K_boot_i.

NOTE Bootstrap key derivation: PRK_boot = HKDF-Extract(salt = RANDOM_CHALLENGE, IKM = PUF_response_hash), K_boot_i = HKDF-Expand(PRK_boot, info = 'CAPSTONE-V1/BOOTSTRAP-KEY' || TARGET_ECU_ID || SESSION_ID). Note: ECU computes PUF_response_hash = SHA-256(PUF_response) to match the enrolled hash stored at CISG. K_i is NOT derived directly from PUF; K_i is a fresh random key generated by CISG post-attestation.

Byte Range	Field Name	Size	Description
Byte 0	PROTOCOL_VER_TYPE	1 B	Version 0x1, Msg Type 0x4 (Key Delivery)
Byte 1	TARGET_ECU_ID	1 B	ECU this key is addressed to
Byte 2	SESSION_ID	1 B	Attestation session this key belongs to
Byte 3	KEY_DERIVATION_ID	1 B	HKDF context label: 0x01 = Session Key
Bytes 4–19	PADDING	16 B	0x00 — reserved (formerly HKDF_SALT, removed to fix circularity)
Bytes 20–35	WRAPPED_K_i	16 B	Ascon-AEAD128 ciphertext of K_i (16B plaintext → 16B ciphertext; tag is separate)
Bytes 36–47	EXPLICIT_NONCE	12 B	Explicit 96-bit nonce portion (see nonce construction below)
Bytes 48–63	KEY_AUTH_TAG	16 B	Ascon-AEAD128 authentication tag for this key delivery frame

⚠ **WARNING** Nonce construction (128 bits = 16 bytes total): SESSION_ID (1B) || TARGET_ECU_ID (1B) || KEY_DERIVATION_ID (1B) || 0x00 (1B) || EXPLICIT_NONCE (12B). This correctly fills Ascon's 128-bit nonce register. Using only 12 bytes (as in v1.0) is a correctness failure — Ascon-AEAD128 requires exactly 128-bit nonces.

3.6 Category 4: Gateway IDS Feature Vector Mapping (Tapan's Pipeline)

After decryption and tag verification, the CISG extracts a 12-dimensional feature vector from each Category 1 frame and feeds it to the Phase 2 ONNX LSTM Autoencoder.

⚠ **WARNING** Failed-Tag Handling Policy (f_6 = 0): If Ascon tag verification fails (f_6 = 0), the CISG drops the frame and zero-fills signal features f_7..f_12 = [0.0, 0.0, 0.0, 0.0, 0.0, 0.0]. This isolates downstream LSTM model state from corrupted or malicious ciphertext while providing a distinct anomaly trigger footprint (f_6 = 0 with potential f_2 counter gap).

Feature Index	Field	Source	Type
f_1	Δt (inter-arrival time)	CISG system clock	float32 (ms)
f_2	Counter_Gap	MONOTONIC_COUNTER delta	uint32
f_3	ECU_ID	SENDER_ECU_ID (Byte 1)	uint8
f_4	Domain_ID	DOMAIN_ID (Byte 2)	uint8
f_5	Msg_Type	Bits 3–0 of Byte 0	uint8
f_6	Tag_Verification_Status	Ascon tag check result	binary (0/1)
f_7	$Signal_1$ — Vehicle_Speed	Decrypted payload bytes 0–3	float32
f_8	$Signal_2$ — Engine_RPM	Decrypted payload bytes 4–7	float32
f_9	$Signal_3$ — Throttle_Position	Decrypted payload bytes 8–11	float32
f_{10}	$Signal_4$ — Brake_Pressure	Decrypted payload bytes 12–15	float32
f_{11}	$Signal_5$ — Steering_Angle	Decrypted payload bytes 16–19	float32
f_{12}	$Signal_6$ — Lateral_Accel	Decrypted payload bytes 20–23	float32

4. Resolved Design Decisions

Decision	Resolution	Rationale
Frame layout	8B header + 40B encrypted payload + 16B tag	8B AD header; full 16B tag; 40B fits 12 signals at float32/uint16 mix
Monotonic counter size	32-bit (4 bytes)	At 10 ms cadence = 497-day rollover. Re-attestation resets counter far more often.
Telemetry signal layout	Float32 mixed (7×f32 + 2×u16 + 1×i16 + 1×u16 + 1×u32 = 40B)	Fills 40B exactly. Scaled-int repacking adds complexity with no Phase 1 benefit.
Key topology	Pure Star — one K_i per ECU, all frames through CISG	Reduces blast radius vs domain keys (N keys not N^2 mesh). Gateway-mediated, same pattern as TLS-terminating proxy.
Byte 3 Bit 5	RESERVED (KEY_TIER dropped from v1.1)	Pure Star has only one key type per ECU; no selection needed.
CISG key selector	Always SENDER_ECU_ID (Byte 1) → K_i	DOMAIN_ID is routing/IDS metadata only; not crypto-relevant in Pure Star.
Category 2 frame count	41 frames — ML-DSA signature only (~2420B ÷ 60B)	ML-DSA public key provisioned at PUF enrollment; not retransmitted at boot.
Category 3.3 payload	Delivers one K_i per ECU	No domain key, no SK_Router. Simplifies Kunal's HKDF to one chain per ECU.

Category 3.3 nonce	16B = SESSION_ID||TARGET_ECU_ID||KEY_DERIV_ID||0x00 + 12B explicit nonce	Correct Ascon-AEAD128 nonce size. 12-byte-only construction (v1.0) was a correctness failure.

5. Notes to Team Members

5.1 Note to Kunal (Cryptography)

NOTE Cryptographic interface frozen for v1.5: (1) Pre-hash PUF response: PUF_response_hash = SHA-256(PUF_response); (2) Derive bootstrap key K_boot_i = HKDF-Extract/Expand using RANDOM_CHALLENGE as salt; (3) Perform single Ascon-AEAD128 protection over 2420B ML-DSA-44 signature -> 2436B protected object (16B tag at end); (4) After verifying attestation signature, generate fresh random 128-bit session key K_i; (5) Wrap K_i under K_boot_i using Ascon-AEAD128 in Category 3.3; (6) Use K_i for Category 1 runtime AEAD.

5.2 Note to Kaustuv (Transport Protocol)

NOTE Category 2 carries ML-DSA-44 SIGNATURE ONLY (not public key). Total size ~2420 bytes -> exactly 41 frames at 60B/frame. Last frame (index 40) is zero-padded with 24 bytes. TOTAL_FRAMES field = 0x29. Category 2 frames are encrypted under Ephemeral Bootstrap Key K_boot_i (Category 3.2 NACK is plaintext). derived immediately after Challenge, NOT runtime key K_i. The MISSING_BITMAP in Category 3.2 is 8 bytes (64-bit), covering frame indices 0–63.

5.3 Note to Tapan (ML / Data Architecture)

NOTE DOMAIN_ID (Byte 2) remains in the AD and in the IDS feature vector (f_4) — it is routing and behavioural metadata, not a crypto key selector. The CISG extracts the 12D feature vector AFTER decryption and tag verification. Tag_Verification_Status (f_6) = 0 means the frame failed authentication — this is itself an anomaly signal and should be treated as a high-weight feature in the LSTM model.

6. Verification Plan

6.1 Automated Tests

• Python test suite (test_frame_parsing.py) using struct/ctypes to pack/unpack all frame categories and verify byte offsets and field sizes.

• Ascon-AEAD128 encrypt/decrypt round-trip tests against NIST SP 800-232 Known Answer Test (KAT) vectors.

• 60/4 block-transfer reassembly simulation with injected missing frames and NACK bitmap verification.

• Nonce uniqueness check: assert no two frames from the same ECU in a session share a nonce (counter monotonicity test).

6.2 Manual Verification

• Cross-check byte layout against Physical_CANFD_Architecture_Specification_v1.1.docx and Trust_Boundary_Analysis_Final_Integrated.docx.

• Confirm Kunal's HKDF parameters match the RANDOM_CHALLENGE usage for bootstrap key derivation.

• Phase 4: HIL timing validation on MCP2518FD hardware to measure CISG decrypt-inspect-re-encrypt latency for same-domain multi-consumer signals.

7. Document History

Version	Date	Change Summary
v1.0	Aug 2026	Initial draft. 4 frame categories. Open questions on counter size and signal layout.
v1.1	Aug 2026	Claude Sonnet 5.0 review: Two-Tier key model (K_session + SK_Router_i). Fixed Category 3.3 nonce to 16B. Fixed Category 2 to signature-only, 41 frames. KEY_TIER bit added to Byte 3.
v1.2	Aug 2026	Claude Sonnet 5.0 re-review: Reverted to Pure Star (one K_i per ECU). KEY_TIER bit dropped (RESERVED). Category 3.3 simplified to single K_i. Same-domain multi-consumer bottleneck documented as Phase 1 known simplification. Citation integrity notice added. All v1.1 nonce/counter/frame-count fixes retained.
v1.3	Aug 2026	Claude Sonnet 5.0 discrepancy review: (1) All Kyber/KEM references removed -- K_i delivery uses Ascon-AEAD128 under attestation-derived bootstrap key; (2) SESSION_ID restored to Category 3.1 Byte 2 (ALGO_SELECTION moved to Byte 3); (3) Category 1 nonce formula added explicitly; (4) ABS removed from Powertrain domain (moved to Chassis as Brake Controller).
v1.4	Aug 2026	Protocol & Spec Discrepancy Fixes: (1) Softened frame length preamble (64B DLC 15 vs 16B DLC 10 NACK); (2) Fixed boot circularity: Cat 2 & 3.2 use Ephemeral Bootstrap Key K_boot_i, NOT K_i; (3) Specified SHA-256(PUF_response) pre-hashing at ECU to match CISG enrollment hash format; (4) Corrected ML-DSA parameter to ML-DSA-44 (FIPS 204 / Dilithium2); (5) Specified zero-fill payload behavior for IDS feature vector on tag failure (f_6 = 0).
v1.5	Aug 2026	Frozen Wire & Crypto Spec: (1) Standardized ML-DSA-44; (2) Whole-object AEAD protection of 2420B signature -> 2436B protected object -> 41 x 60B transport frames; (3) One fresh 16B nonce per attestation transfer; retransmissions resend original fragment; (4) Key lifecycle frozen: K_boot_i from pre-hashed PUF + challenge; fresh random 128-bit K_i post-attestation; (5) Category 3.2 NACK is boot-time Category-2-only; Category 1 telemetry is NEVER retransmitted; (6) Category 2 payload renamed to 'fragment of the Ascon-AEAD128 protected attestation object'.
v1.6	Aug 2026	Final Adjustments: (1) Cat-2 nonce uses RANDOM_CHALLENGE directly; (2) Corrected padding arithmetic (36 real + 24 padding) and Dilithium2 references; (3) Replaced FOB with CISG secure enrollment database; (4) Clarified Cat 2 transport metadata is not AD.